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The role of shelterwood cutting and protection against game browsing for the regeneration of silver fir

Die Rolle des Schirmschlages und des Wildverbisschutzes für die natürliche Verjüngung der Weißtanne

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Jaroslav Simon³, Václav Štícha¹

Key words: natural mixed forest, silver fir (*Abies alba* Mill.), European beech (*Fagus sylvatica* L.), Norway spruce (*Picea abies* /L./ Karst.), biodiversity, fir decline, Giant Mountains

Stichworte: natürlicher Mischwald, Weißtanne (*Abies alba* Mill.), Rotbuche (*Fagus sylvatica* L.), Fichte (*Picea abies* /L./ Karst.), biologische Vielfalt, Tannerückgang, Riesengebirge

Abstract

This paper describes forest stands on four permanent research plots (PRP) situated in the National Park of Krkonoše (Czech Republic) and Karkonoski Park Narodowy (Poland). The aim of this paper was to evaluate basic forest stand characteristics with emphasis on natural regeneration of silver (*Abies alba* Mill.) as result of active forest management in the past decades. The horizontal structure of the overstorey was mostly random and that of natural regeneration was aggregated. The vertical structure was distinctly more differentiated in forest stands with largely advanced regeneration compared to stands in the initial stage of regeneration. This mixed near-natural

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forests with silver fir on the studied PRP were characterized by the significantly differentiated structure that is connected with long-term anthropogenic influences. The standing volume of the main and secondary stand on the particular PRP was in the range of 237-598 m³.ha⁻¹. It has been documented on the studied plots that hoofed game had a very negative or even destructive influence on fir natural regeneration. Distinct stand variability was indicated by all used structural indices, also species richness on the particular plots showed high variability, the characteristics of stand density indicated relatively high diversity and complexity of the structure of mixed fir stands on all PRP.

Zusammenfassung

Dieser Artikel beschreibt Bestände auf vier dauerhaften Forschungsflächen, die sich im Nationalpark Krkonoše (Riesengebirge, Tschechische Republik) und im Nationalpark Karkonoski Park Narodowy (Polen) befinden. Die vorliegende Arbeit setzt sich zum Ziel, grundlegende Charakteristiken des Bestandes auszuwerten. Die horizontale Struktur der oberen Etage ist in der Regel zufällig. Dagegen ist die Naturverjüngung aggregiert verteilt. Die vertikale Struktur zeigt höhere Differenzierung, wo die Phase des Nachwachsens der Naturverjüngung fortgeschritten ist, im Vergleich mit Beständen, wo sich die Verjüngung in ihrer Anfangsphase befindet. Diese gemischten naturnahen Wälder mit Anteil der Weißtanne sind auf den studierten dauerhaften Forschungsflächen durch eine differenzierte Struktur gekennzeichnet, die mit einer langfristigen Einwirkung von anthropogenen Faktoren verbunden ist. Das Volumen des Haupt- und Nebenbestandes auf einzelnen dauerhaften Forschungsflächen bewegt sich zwischen 237 und 598 m³.ha⁻¹. Auf studierten Flächen wurde dokumentiert, dass Schalenwild eine negative oder sogar destruktive Auswirkung auf die Naturverjüngung der Weißtanne hat. Die unterschiedliche Bestandsvariabilität ist durch den Variabilitätsindex belegt. Die Struktur- und Artenvielfalt auf einzelnen Forschungsflächen weist eine große Variabilität auf, die Charakteristiken der Mischbestände auf dauerhaften Forschungsflächen zeigen insgesamt hohe Vielfalt und Komplexität der Struktur.

1. Introduction and problem analysis

Mixed fir forests belong to the main ecosystems in Europe (LEBOURGEOIS et al. 2013). For these forests complicated structure and development dynamics are characteristic (KORPEL 1995), thus close-to-nature forest management with emphasis on natural regeneration and selection principles seems to be the best choice for these stand types (PUETTMANN et al. 2009). In Europe the silver fir (*Abies alba* Mill.) occurrence is mostly situated to mountain forests, preferably to localities with colder climatic conditions, low temperature amplitudes and relatively high annual precipitation (ROBAKOWSKI et al. 2004). The fir occurs scarcely in lowland forests (DOBROWOLSKA 1998). In the Krkonoše Mts. the mixed fir stands are characterized by a long, 350-400-year development period (VACEK et al. 2010a), from the Carpathian Mts. The range of 400-450 years was reported (KORPEL 1995). In the Krkonoše Mts. the forests were distinctly influenced in the past at the cost of particularly fir, and therefore the study and protection of remaining mixed fir forests are very important (SCHWARZ 1997). Taking into account climate changes and extreme climatic phenomena that have already affected the species distribution and growth of single trees in many localities all over Europe (LENOIR et al. 2008; WAY, OREN 2010), the fir shows higher vulnerability to summer drought at lower locations and to low temperatures at higher locations (LEBOURGEOIS et al. 2013).

The process of regeneration is a highly stochastic phenomenon of forest dynamics, depending on a larger scale on many factors that include habitat, stand characteristics, history of land use or the influence of herbivores (PALUCH 2005). Spatial and temporal interactions between species are largely dynamic, being variable in relation to soil, climatic and stand conditions (FORRESTER et al. 2013). Silver fir is regarded as the most shade tolerant commercial tree species in European forests (STANCIOIU, O'HARA 2006). On rich sites this tolerance to shade can be even higher (KUČERAVÁ et al. 2013), nevertheless light intensity is the only growth factor that can be influenced directly by silvicultural measures (NICOTRA et al. 1999, DOBROWOLSKA, VEBLEN 2008). In accordance with many authors (KANTOR 2001, STANCIOIU, O'HARA 2006; MUSCOLO et al. 2010) for the natural regeneration of silver fir the protection of parent stand against drought and frost is of crucial importance.

Fir regeneration in mixed forests of the Krkonoše Mts. takes place almost exclusively under the shelterwood of parent stand, fir tends to regenerate in clumps like spruce while beech regenerates on larger areas (VACEK et al. 2010b). In the last decades there has been a decrease in the fir share whereas an increase in the beech share has been relatively distinct (JAWORSKI et al. 2002, FIRM et al. 2009). A pronounced decrease in the fir share is caused by several factors. One of the most important factors in conditions of the Krkonoše Mts. is high food attractiveness of fir to hoofed game, especially to red and roe deer (SCHWARZ et al. 2007). Fir regeneration is also limited by increased vulnerability to frost and drought (CLOSSET-KOPP et al. 2006).

With the increased population levels of game its excessive activity in forests is a factor that was proved to suppress forest undergrowth diversity and tree layer regeneration, namely of highly palatable tree species. The attractiveness of the species is highly specific and it increases in case that attractive species (e.g. *Abies alba*, *Acer* spp.) occur in combination with less attractive ones (e.g. *Picea abies*) (AMMER 1996, SENN, SUTER 2002). Sporadic occurrence and, on the contrary, its dominant position are often cited as factors influencing food attractiveness. In the overwhelming majority of cases the food attractiveness of a tree species increases with a decreasing share of tree species in natural regeneration (AMMER 1996, ČERMÁK et al. 2009). Considering the present share of fir in the Krkonoše Mts. (0.1%), the very high food attractiveness of fir is obvious in these areas. Natural regeneration of fir is very difficult there and it is practically impossible in most localities outside game-proof fenced enclosures or without individual protection (SCHWARZ 1997).

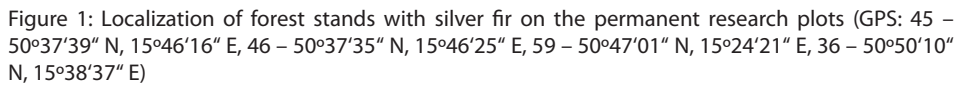
The present paper evaluates the structure and natural regeneration in forest stands in one of the most valuable natural mixed fir forest remnants in the Central Sudetes of the Czech and Polish part of the Krkonoše (Giant) Mts. The aim of this study was (1) to quantify the impact of ungulates on the natural regeneration of fir, on the height structure of regeneration and on the shift of tree species composition and (2) to analyse the effect of silvicultural measures on parents stand and fir regeneration in genetically valuable areas, all as background information for the definition of near-natural management in similar site conditions.

2. Material and Methods

2.1. Study area

The paper was conducted on four permanent research plots (PRP) situated in the National Parks of Krkonoše Mts. The parent rock of these parks is formed mainly by granite, mica schist and phyllite. At the lowest altitudes Cambisols are dominant and above 1 000 m Podzols prevail. Average precipitation varies with respect to altitude from 650 mm to 1260 mm and mean annual temperature decreases with altitude from 7.6 °C to 2.6 °C.

Fig. 1 shows the localization of PRP and Tab. 1 contains an overview of basic data on PRP. PRP 45, 46 and 59 are situated in the Krkonoše National Park on the Czech side of the mountain range and PRP 36 lies in Karkonoski Park Narodowy on the Polish side of the mountain range.



	Name of PRP	Altitude (m)	Exposition	Slope (°)	Forest site type	Species	Age	Diameter (cm)	Height ₉₅ (m)	Volume (m³.ha⁻¹)
45	Jánské Lázně	730	NE	20	6K1	AA, PA, FS, BP, SA, AP, UG	109	15.4	22.8	302
46	Jánské Lázně	670	SW	18	6K1	AA, PA, FS, PT, AP, AG, FE, SA	108	34.9	37.6	598
59	Nad Černou ručejí	710	SW	15	6K1	AA, PA, FS, BP, SA	126	20.5	32.5	237
36	Chojník	520	N	16	4S1	AA, PA, FS, LD, AP, API	124	34.5	35.2	591

Notes: Age – mean age of tree layer, Height₉₅ – 95% quantile of heights, Forest site type: 4S – *Fagetum oligo-mesotrophicum*, 6K – *Piceeto-Fagetum acidophilum* (both modal cambisol); Species: AA – *Abies alba*, PA – *Picea abies*, FS – *Fagus sylvatica*, BP – *Betula pendula*, SA – *Sorbus aucuparia*, AP – *Acer pseudoplatanus*, UG – *Ulmus glabra*, PT – *Populus tremula*, AG – *Alnus glutinosa*, FE – *Fraxinus excelsior*. LD – *Larix decidua*, API – *Acer platanoides*.

PRP 45 and 46 were established in 2013 in the core zone of the Janské Lázně genetic resources area designed for silver fir. The initial site and stand conditions were comparable on both PRP before the beginning of regeneration. Unlike PRP 46, PRP 45 was fenced in the preceding two decades to provide for natural regeneration of fir. In the past 20 years, on PRP 45 were realized three shelterwood cuttings (preparatory, seeding and liberation cuttings). On PRP 46 only preparatory cutting and seeding cutting with low intensity were carried out. Fir regeneration here were treated chemically against browsing.

PRP 59 was established in 2013 and is protected by fencing. Beside this plot, damage caused by game was studied in the same stand on an area of 0.25 ha in the same way outside the game-proof fenced enclosure. In the past 20 years, on PRP 59 were also realized three shelterwood cuttings. Same type of cuttings were applied outside the fence, where fir regeneration were treated chemically.

PRP 36 was established in 2004 and is not protected by fencing. For the observation of the growth of recruits without the influence of hoofed game, dynamics of the growth of recruits was studied in the same stand in a small fenced enclosure beside this plot. This plot is left to develop without intervention. Table 2 shows information about silvicultural measures and harvest intensity for particular forest stands in study.

Table 2: Overview of shelterwood felling on the permanent research plots for forest stand

PRP	Fenced (period)	Stand volume before harvest	Preparatory felling year / m ³ .ha ⁻¹	Seeding felling year / m ³ .ha ⁻¹	Liberation felling year / m ³ .ha ⁻¹	Final felling year / m ³ .ha ⁻¹
45	yes (1991-2005)	672	1986 / 65	1990 / 152	2000 / 245	not planned
46	no	661	2008 / 67	–	–	not planned
59	yes (2001-now)	672	1986 / 75	1990 / 186	2000 / 289	not planned
36	no	591	–	–	–	not planned
control to 36	yes (2002-now)	620	2001 / 240	–	–	not planned

2.2. Data collection

The FieldMap technology (IFER-Monitoring and Mapping Solutions Ltd.) was used for the establishment and measurements 50 × 50 m (0.25 ha) plots. The position of all overstorey trees was localized by this technology and crown projections were also measured in the tree layer at ≥4 points along the crown perimeter. The diameters at breast height (dbh ≥ 4 cm) of trees were measured with a metal calliper (accuracy 1 mm) and tree heights with height to the base of the live crown were measured with a laser Vertex (accuracy 0.1 m). A Pressler auger was used to collect samples from trees at the breast height to determine age and increment.

In natural regeneration in all measured recruits (from height ≥ 5 cm to dbh < 4 cm) on the particular PRP crown height and width were measured with a height stick and tape (accuracy 1 cm). In all recruits broken down by tree species browsing of terminal apex, lateral browsing and repeated browsing were investigated on PRP and in their environs. In juvenile stages of silver fir in the fenced enclosure and outside the enclosure in the immediate proximity of the studied PRP sample trees of typical recruits were taken for growth analyses, 5 sample trees from each variant.

2.3. Data processing

The following indices were used to evaluate the stand diversity in relation to tree species composition, abundance, horizontal and vertical distribution: Arten-profil index (PRETZSCH 2006), diameter differentiation index (FÜLDNER 1995), species richness index (MARFALEF 1958), species heterogeneity index (SHANNON 1948), species evenness index (PIELOU 1975), stand diversity index (JAEHNE, DOHRENBUSCH 1997), index of non-randomness (PIELOU 1959, MOUNTFORD 1961) and L -functions (RIPLEY 1981). The criteria for these indices are given in Table 3. For a completely random distribution of points $L(r) = \pi r^2$ is applicable. In case that the value $L(r)$ calculated from real data is higher than the expected value, it can be explained as a trend of forming point clusters at a distance r . Inversely, the lower $L(r)$ value indicates a trend of repulsion, i.e. the formation of regular point structures. The calculation of these characteristics was made using the growth simulator SIBYLA (FABRIKA, ĎURSKÝ 2005) and PointPro 2.1 software (Zahradník).

Table 3: Overview of indices describing the stand structure and their common interpretation

Criterion	Quantifiers	Label	Reference	Evaluation
Vertical diversity	Arten-profil index	A (Pri)	PRETZSCH 2006	range 0-1; balanced vertical structure $A < 0.3$; selection forest $A > 0.9$
Structure differentiation	Diameter differentiation	TMd (Fi)	FÜLDNER 1995	range 0-1; low TM < 0.3 ; very high differentiation TM > 0.7
Horizontal structure	Index of non-randomness	α (P&Mi)	PIELOU 1959, MOUNTFORD 1961	mean value $\alpha = 1$; aggregation $\alpha > 1$; regularity $\alpha < 1$
Species diversity	Species richness	D (Mi)	MARFALEF 1958	minimum D = 0, higher D = higher values
	Species heterogeneity	H' (Si)	SHANNON 1948	range 0-1; minimum H' = 0, maximum H' = 1 – for 10 tree species (Hertley measure)
	Species evenness	E (Pii)	PIELOU 1975	range 0-1; minimum E = 0, maximum E = 1
Complex diversity	Stand diversity	B (J&Di)	JAEHNE, DOHRENBUSCH 1997	monotonous structure B < 4 ; uneven structure B = 6-8; very diverse structure B > 9

The tree volume was calculated according to the volume equations published in PETRÁŠ, PAJTÍK (1991). As for the density characteristic, the relative stand density in-

dex (SDI; REINEKE 1933) and crown projection area were observed for each plot. The layout maps were made in the ArcGIS software (Esri).

Analyses were processed in the STATISTICA 12 software (StatSoft, Tulsa). Data were log transformed to acquire normal distribution and tested by using Kolomogorov-Smirnov test. The difference in height of natural regeneration between research plots and species and between current height growth compared to reduced growth caused by browsing were separately tested by one-way analysis of variance (ANOVA). Significant differences were subsequently tested by post-hoc comparison Tukey's HSD tests. Significance of statistic was noted as follow: $p > 0.05$, $p < 0.05$, $p < 0.01$ and $p < 0.001$.

3. Results

3.1. Species composition

Tab. 4 shows the species composition of tree layer compare with natural regeneration. The representation of particular tree species on the plots was largely different. The highest share of fir was on PRP 59 (38%), where beech did not practically occur, while spruce was dominant species on this plots and its share amounted to 59% there. On the remaining plots beech was dominant with the share of about 45%; a high share of fir was also found out on PRP 36 (36%), on the other plots its share was lower at the cost of spruce. The presence of the other tree species (*Betula pendula*, *Sorbous aucuparia*, *Acer pseudoplatanus*, *Ulmus glabra*, *Populus tremula*, *Alnus glutinosa*, *Fraxinus excelsior*, *Larix decidua*, *Acer platanoides*) was only sporadic with the highest ratio on PRP 46 (6.9 %) and they were interspersed as single trees.

Table 4: Overview of species of tree layer and natural regeneration on the permanent research plots

PRP	Tree layer					Natural regeneration				
	(% of volume)				(trees.ha ⁻¹)	(% of individuals)				(pcs.ha ⁻¹)
	Fir	Spruce	Beech	Others		Fir	Spruce	Beech	Others	
45	15.9	36.8	47.4	0.0	1 624	0.3	30.8	66.1	2.8	156 800
46	21.6	28.4	43.1	6.9	456	20.2	10.9	61.3	7.6	23 800
59	38.0	59.1	3.0	0.0	628	25.9	27.8	42.1	4.2	30 900
36	35.7	17.9	44.5	1.9	464	5.6	0.7	90.8	2.9	90 080

3.2. Tree layer growth characteristics

The stand characteristic on the particular PRPs is documented in Tab. 5. The standing volume on PRP 45 was 302 m³.ha⁻¹ (fir 48 m³, spruce 111 m³ and beech 143 m³) with

the highest numbers of trees among all PRPs (total number of 1 625 trees.ha⁻¹). Periodic annual increment on PRP 45 was 6.5 m³.ha⁻¹.y⁻¹ and mean annual increment 2.77 m³.ha⁻¹.y⁻¹. The highest standing volume was on PRP 46 - 598 m³.ha⁻¹ (fir 129 m³, spruce 170 m³ and beech 258 m³) with 456 trees.ha⁻¹. Periodic annual increment on PRP 46 was 8.8 m³.ha⁻¹.y⁻¹ and mean annual increment 5.59 m³.ha⁻¹.y⁻¹. The lowest stand volume was on PRP 59 - 237 m³ (fir 90 m³, spruce 140 m³ and beech 7 m³) with 628 trees.ha⁻¹. Periodic annual increment on PRP 59 was 4.6 m³.ha⁻¹.y⁻¹ and mean annual increment 1.88 m³.ha⁻¹.y⁻¹. On PRP 36 the standing volume was 591 m³.ha⁻¹ (fir 211 m³, spruce 106 m³ and beech 263 m³) with 464 trees.ha⁻¹. Periodic annual increment on PRP 36 was 8.65 m³.ha⁻¹.y⁻¹ with mean annual increment 4.77 m³.ha⁻¹.y⁻¹.

Table 5: Basic stand characteristic on the permanent research plots (2014)

PRP	t (y)	dbh±SD (cm)	h (m)	h ₉₅ (m)	f	v (m ³ .trees ⁻¹)	N (trees.ha ⁻¹)	G (m ² .ha ⁻¹)	V (m ³ .ha ⁻¹)	h:d	PAI (m ³ .ha ⁻¹ .y ⁻¹)	MAI (m ³ .ha ⁻¹ .y ⁻¹)
FIR												
45	80	10.7±4.8	7.81	11.3	0.68	0.048	992	8.9	48	73.0	1.95	0.60
46	104	41.6±10.6	31.59	35.4	0.44	1.902	68	9.2	129	75.9	1.95	1.24
36	124	37.4±9.5	28.15	32.9	0.45	1.385	152	16.7	211	75.3	3.40	1.70
59	130	25.0±21.5	7.22	32.6	1.68	0.595	152	7.3	90	28.9	1.50	0.69
SPRUCE												
45	110	19.2±13.1	11.09	33.0	0.95	0.306	364	10.4	111	57.8	2.15	1.01
46	118	60.9±7.1	38.20	40.2	0.38	4.256	40	11.6	170	62.7	1.95	1.51
59	127	28.7±19.4	15.26	34.9	0.78	0.776	180	11.6	140	53.2	2.55	1.10
36	118	42.5±8.9	31.02	36.3	0.40	1.760	60	8.5	106	73.0	1.65	0.90
BEECH												
45	119	24.4±11.4	11.42	29.0	1.14	0.607	236	10.9	143	46.8	2.40	1.20
46	107	30.4±13.0	25.24	32.0	0.54	0.991	260	18.9	258	83.0	4.40	2.41
59	63	9.0±4.6	6.02	10.8	0.66	0.026	272	1.7	7	66.9	0.50	0.11
36	128	33.0±22.3	17.37	35.7	0.89	1.315	200	16.9	263	52.6	3.35	2.05
TOTAL STAND												
45	109	15.4±10.4	9.07	22.8	1.10	0.186	1 624	30.3	302	58.9	6.50	2.77
46	108	34.9±16.9	25.35	37.6	0.54	1.311	456	43.5	598	72.6	8.80	5.59
59	126	20.5±16.1	20.50	32.5	1.27	0.377	628	20.6	237	43.7	4.55	1.88
36	124	34.5±18.4	22.22	35.2	0.61	1.274	464	43.3	591	64.4	8.65	4.77
Notes: PRP – permanent research plot, t – average age of stand, dbh±SD – average breast height diameter ± standard deviation, h – mean stand height, h ₉₅ – top height – 95% quantile, f – form factor, v – mean tree volume, N – number of trees, G – stand basal area, V – stand volume, h:d – slenderness ratio, PAI – periodic annual increment, MAI – mean annual increment												

Histograms of diameter classes in Fig. 2 document a very distinct variability in the representation of trees in diameter classes on PRPs. The highest representation of fir trees in low diameter classes was found out on PRP 45, where fir was a highly dominant species. Relatively high representation of fir in low diameter classes was found also on PRP 59. However, on this plot the representation of spruce and also mainly of beech was high; the latter species was distinctly dominant in low diameter classes on

PRP 36 and 46. On all PRPs there were trees in high diameter classes and their abundance corresponds with the numbers of trees in lower classes.

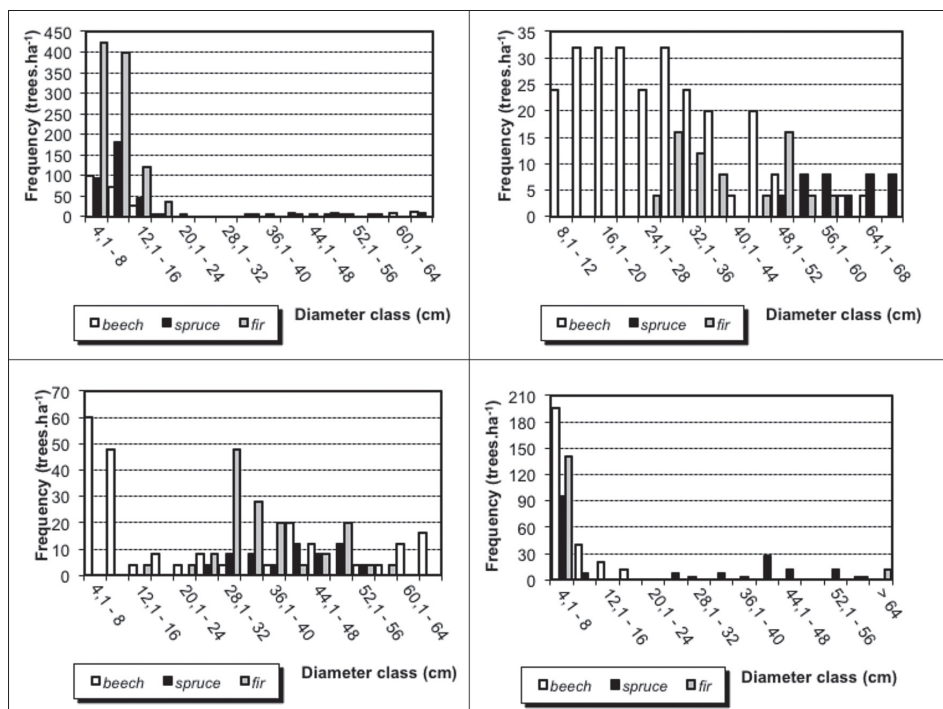


Figure 2: Frequency of diameter classes of tree layer per hectare on the permanent research plots broken down by main tree species

The characteristics of stand density on PRP 45, 46 and 36 maintained a high level (SDI 0.66-0.71) while on PRP 59 the values were distinctly lower (SDI 0.37). As for the crown contact on PRP 45, 46 and 36, the crowns largely interpenetrated and influenced each other. The total area of crown projections was currently in the range of 1.0 (PRP 59) to 3.1 ha (PRP 46).

3.3. Tree layer biodiversity

Fig. 3 illustrates the horizontal structure of the studied mixed stands with silver fir on PRPs 45, 46, 36 and 59. In this figure more advanced regeneration with a distinctly higher number of recruits in the growing up stage (of the lower tree layer) can be seen on PRP 45 (it was fenced for 15 years) and on PRP 59 (it is fenced) compared to PRP 46 and 36, which are not fenced.

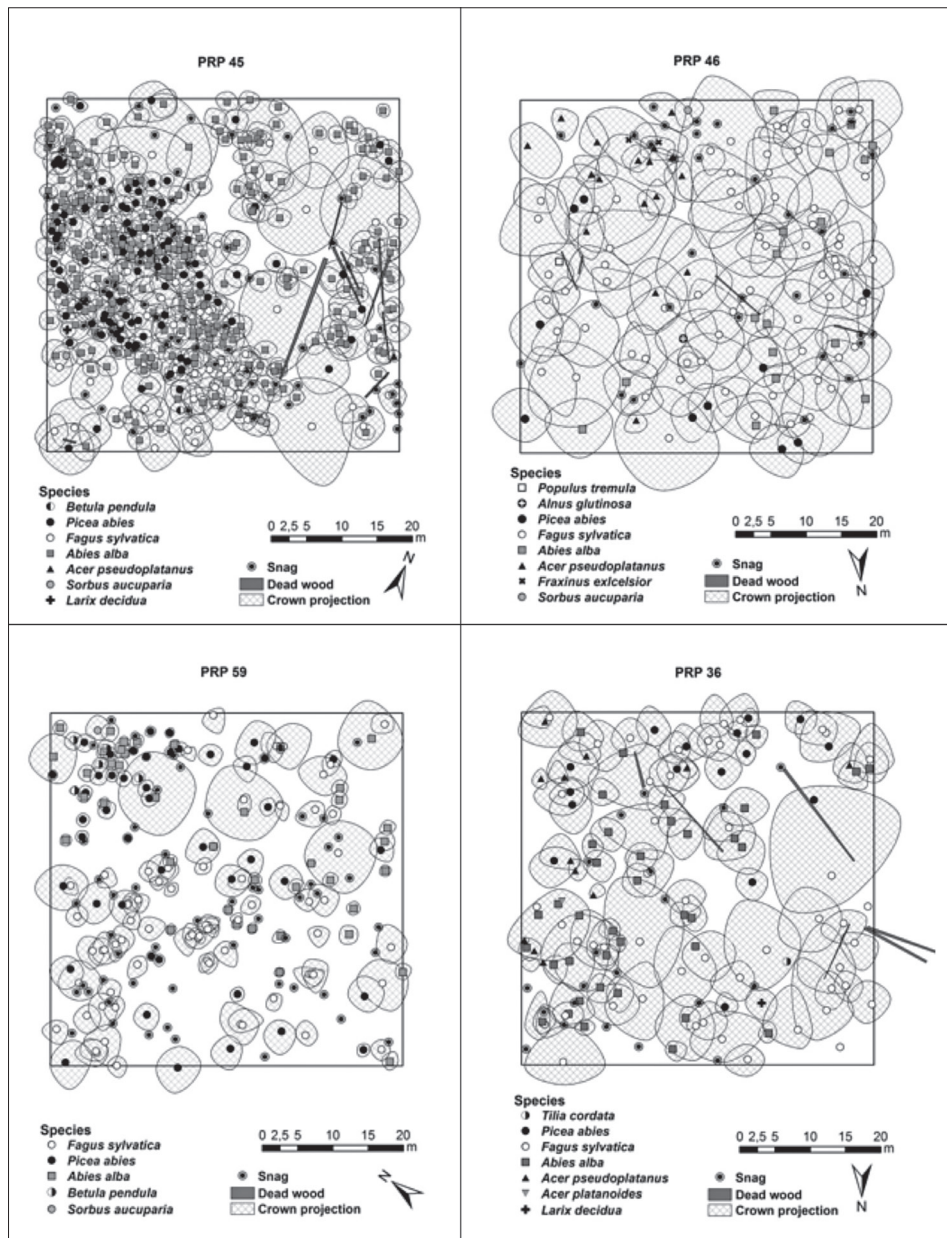


Figure 3: Horizontal structure of forest stands with silver fir on the permanent research plots

The values of structural indices of the overstorey trees on the particular PRP are shown in Tab. 6. According to α index the spatial pattern of all trees on PRP 45 was aggregated. The mostly aggregated distribution of fir trees and also the all tree layer according to their distance (spacing) on PRP 45 was indicated by the L -function (Fig. 4). On PRP 46 spatial pattern of tree layer and all particular tree species was random. Random distribution of tree layer according structural indexes in total and of all particular tree species on 59 was found; only in fir trees was the distribution aggregated according α index. The random distribution of tree layer according to their distance is also indicated by the L -function to a distance of 2 m, for a larger spacing than 2 m the distribution was moderately aggregated. The distribution of fir trees was similar (Fig. 4). On PRP 36 similar spatial pattern of overstory was detected. Only in fir was the distribution of trees aggregated, but the random distribution of fir trees is evident from the L -function to a spacing of 6 m (Fig. 4).

Table 6: Indices describing the biodiversity of tree layer

PRP	A (Pri)	TM _d (Fi)	α (P&Mi)	H' (Si)	E (Pii)	D (Mi)	B (J&Di)
45	0.608	0.280	1.567*	0.487	0.576	0.812	10.511
46	0.563	0.400	1.017	0.591	0.654	1.143	9.204
59	0.489	0.424	1.172	0.399	0.571	0.621	9.294
36	0.604	0.517	1.178	0.512	0.658	0.814	9.338

Notes: A – Arten-profil index, TM_d – diameter differentiation index, TMh – height differentiation index, α – index of non-randomness, H' – species heterogeneity index (entropy H'), E – species evenness index, D – species richness index, B – stand diversity index;
 *statistically significant for horizontal structure (p>0.05)

In terms of the vertical structure, according to the A index the construction of mixed fir stands was medium to high on all PRP, ranging between 0.49 and 0.61. The B index indicated in all plots very high stand diversity in the range of 9.20 to 10.51. Diameter differentiation was largely different on the particular plots; it was the highest on PRP 36, where it reached moderately higher values, while on the other plots diameter differentiation was lower. The lowest diameter differentiation was found out on PRP 45. The forest stands were characterized by higher species diversity and relatively high species evenness when the highest values were reached on PRP 36 and 46. Species richness was very different on the particular plots. PRP 46 and 45 had high species richness, PRP 36 had intermediate richness and PRP 59 low to intermediate richness.

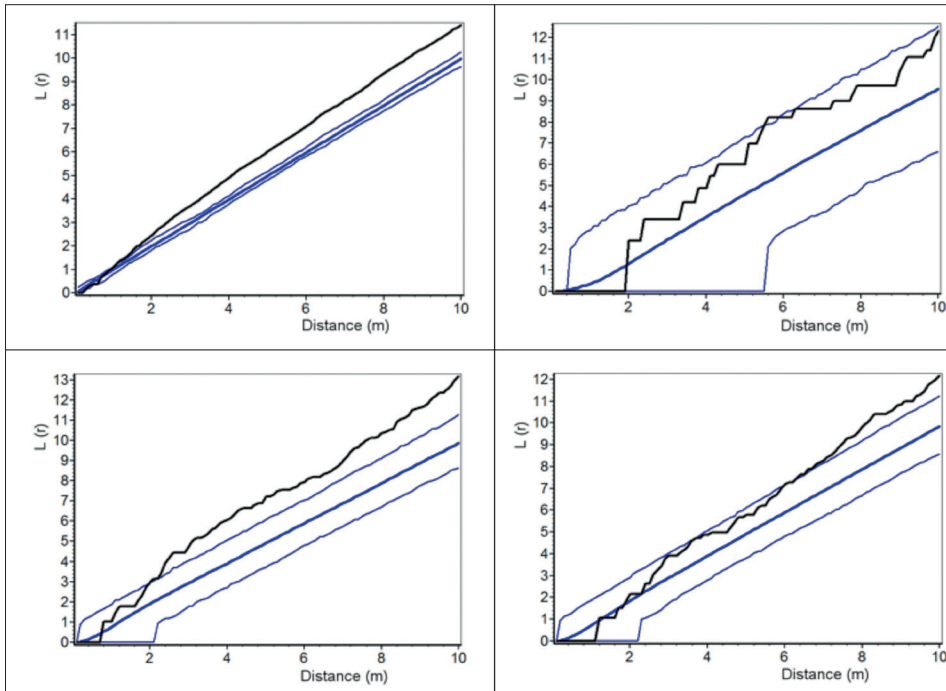


Figure 4: Horizontal structure of the fir tree layer expressed by Ripley's L -function

3.4. Natural regeneration

Obviously, beech had the highest share in natural regeneration on the plots, amounting up to 91% on PRP 36 (Tab. 4). The lowest share of beech was observed on PRP 59 (42%), where the highest share of fir (26%) on the studied PRP was found out. The high share of fir was a result of the intensive protection by fencing there. The richest species composition of natural regeneration was on PRP 36, where seven species are represented although beech was highly dominant there; the poorest species composition was on PRP 45, where only four tree species were found out. In total, the numbers of recruits were high to very high on the given PRP and with the intensive protection against game like on PRP 59 it was possible to reach a distinct share of fir which was prospering there. The distribution of recruits according to tree species was aggregated in all cases and PRPs.

The height structure of natural regeneration of main species on the particular PRP as converted per ha is illustrated in Fig. 5. The figure shows that the height structure on the PRP was distinctly differentiated and the representation in height classes was very variable. On PRP 45 and 59 the situation was very similar and the recruits rather

belong to higher diameter classes. On PRP 45 the histogram of height frequencies had a distinct two-peak shape while there is a considerably downward trend from the peaks in both cases. This two-peak shape was probably caused by rich mast years with good conditions for natural regeneration, especially at the time when the PRP used to be fenced. On PRP 46 and 36 the highest share of recruits belonged to the first height class; on PRP 46, besides this class, the representation was relatively balanced to a height of 120 cm and above this height the share of recruits was low. On PRP 36 there were mostly recruits of maximally 60 cm in height, taller recruits occur on this plot only sporadically. Fig. 5 documents that on the plots that had not been fenced (PRP 46 and 36) the fir mostly occurred in the two lowest height classes.

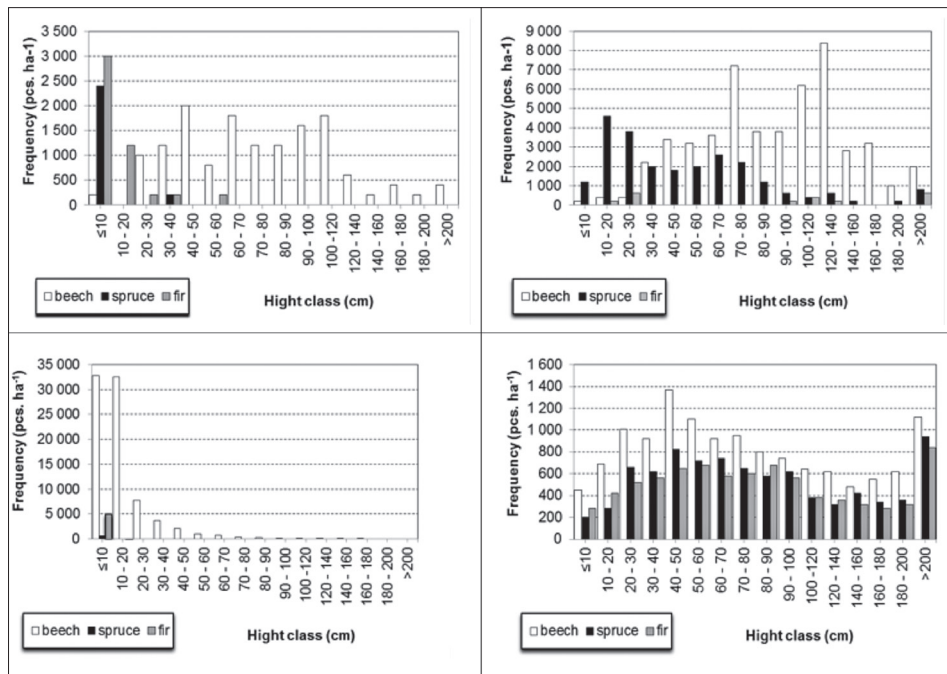


Figure 5: Frequency of height classes of natural regeneration per hectare on the permanent research plots broken down by main tree species

Comparing the mean heights of natural regeneration of fir, spruce and beech among PRPs together, recruits showed significant differences ($F_{(6, 5570)} = 5570$, $p < 0.001$; Fig. 6). On all separately tested PRPs also significant differences in recruit mean heights among tree species were observed (PRP 45, 46 $p < 0.001$; PRP 59, 36 $p < 0.05$). The highest recruits mean heights were found on PRP 59, especially fir ($105.7 \text{ cm} \pm 6.2 \text{ SE}$),

followed by PRP 45, particularly ($103.3 \text{ cm} \pm 2.6 \text{ SE}$). The lowest recruits were found on PRP 36, especially fir ($6.5 \text{ cm} \pm 5.0 \text{ SE}$), followed by PRP 46, particularly spruce ($8.5 \text{ cm} \pm 9.0 \text{ SE}$). The difference was most discernible in fir: e.g. on PRP 45 (protected by fencing for 15 years) the mean height of fir recruits was $105.7 \text{ cm} \pm 6.2 \text{ SE}$ (and almost 984 recruits reached the parameters of the lower tree layer) and on PRP 46 it was only $11.9 \text{ cm} \pm 6.2 \text{ SE}$. A slightly better situation was observed in beech when on PRP 45 the mean height of recruits was $103.2 \text{ cm} \pm 1.5 \text{ SE}$ and on PRP 46 it was only $67.5 \text{ cm} \pm 3.9 \text{ SE}$. In general, it is to state that the plots that have been protected against game by fencing (PRP 45 and 59) and those without protection against game (PRP 46 and 36) are very similar in the trend of the natural regeneration growth. Nevertheless, the fenced stands fir recruits reached the highest mean height, while on unfenced stands the mean height of beech recruits was significantly higher than other tree species ($p < 0.05$).

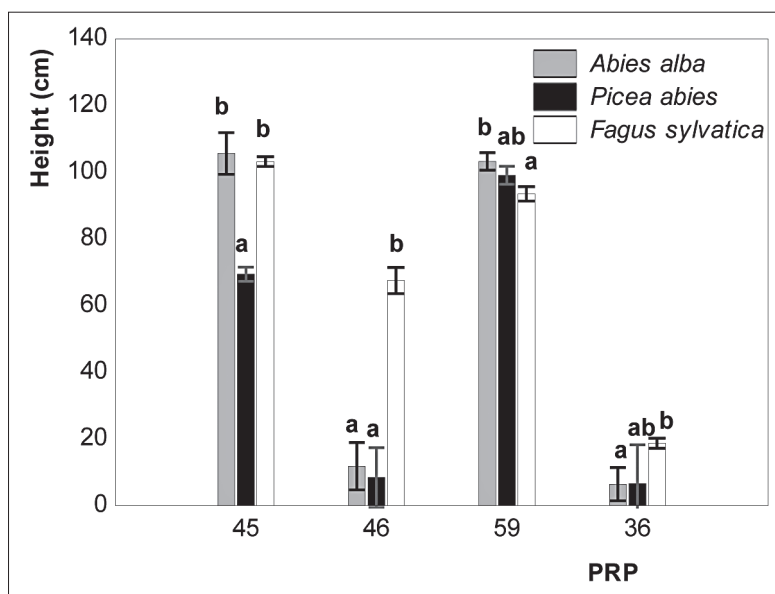


Figure 6: Mean height of natural regeneration recruits on the permanent research plots according to tree species. Significant differences ($p < 0.05$) among tree species on each plot separately are indicated by different letters. Error bars represent standard error of the mean (SE)

3.5. The influence of hoofed game on natural regeneration

Hoofed game had a significant effect on the mean height of natural regeneration recruits of fir ($F_{(1, 525)} = 69.3$, $p < 0.001$). Height of full growth of sample trees in the game-proof fenced enclosure was significantly higher than reduced growth caused

by browsing outside this enclosure (Fig. 9). The figure illustrates that in a group shelterwood regeneration system the studied tree species were not to a larger extent damaged by game in the juvenile stage (until 5 years of age, $F_{(1, 30)} = 8.9$, $p < 0.05$) however, when they exceeded the height of the herb layer, they suffered from high browsing pressure with following suppressed height growth a higher mortality rates. This situation was seen mainly on PRP 36 where no recruits older than 10 years practically occurred at all. The situation on the two remaining plots was very similar, and e.g. at 20 years of silver fir age on PRP 59 the values were $411 \text{ cm} \pm 21.3 \text{ SE}$ in the fenced part and $134 \text{ cm} \pm 10.1 \text{ SE}$ (33%) outside the fenced part influenced by browsing.

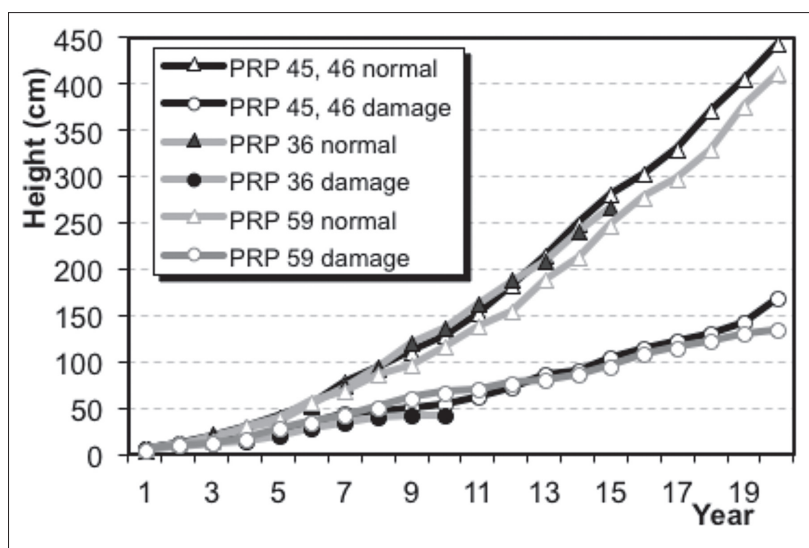


Figure 7: Full height growth (normal) of fir recruits compared to reduced growth (damage) caused by browsing on the permanent research plots

Damage caused by browsing by hoofed game to natural regeneration broken down by PRPs and tree species is shown in Tab. 7. As recruits higher than 15 cm suffered from browsing damage to natural regeneration, damage in this table was calculated in recruits higher than 15 cm. The table documents that the greatest damage by browsing was observed in fir (browsing of the terminal apex in 82-94% of recruits, the exception is PRP 45, which was fenced for 15 years to provide for fir regeneration and where such damage was 22% only). In beech great damage to the terminal apex in the range of 41-44% was found out on PRP 46 and outside the fenced enclosure on PRP 59. On plots with the mass occurrence of beech regeneration damage to the terminal apex caused by game was minimal (4-5%). Damage by browsing to the terminal apex in spruce ranged from 2 to 20%.

Table 7: Damage caused by browsing and fraying by the game to natural regeneration of fir (last two lines of beech and spruce) on the studied unfenced PRPs and on location in close proximity to the fenced PRPs

Type of browsing		PRP 45		PRP 46		PRP 59		PRP 36	
		ind.ha ⁻¹	%	ind.ha ⁻¹	%	ind.ha ⁻¹	%	ind.ha ⁻¹	%
Browsing of the terminal apex	1time	76	6.91	48	4.00	44	61.11	444	21.76
	2-4times	140	12.73	540	45.00	24	33.33	852	41.76
	5times and more	24	2.18	472	39.33	0	0.00	368	18.04
	total	240	21.82	1 060	88.33	68	94.44	1 664	81.56
Lateral browsing		268	24.36	1 024	85.33	72	100.00	1 748	85.67
Fraying		8	0.73	0	0.00	0	0.00	24	1.18
Fir regeneration (ind.ha ⁻¹)		2 200	100.00	4 800	100.00	5 400	100.00	2 280	100.00
Beech – terminal browsing		2 440	4.74	6 280	43.61	1 280	3.66	4 240	40.76
Spruce – terminal browsing		80	0.39	40	20.00	0	0.00	480	7.01
Note: share in % browsing was calculated from recruits ≥ 15 cm									

4. Discussion

Mixed near-natural forests with silver fir on the studied PRPs are characterized by significantly differentiated structure that is connected with long-term anthropogenic influences (e.g. long-term air-pollution load, inappropriate forest management especially in the last centuries, structure and dynamics disturbance by the wildlife population). These findings from spruce-fir-beech stands in the Trčkov National Nature Reserve in the Orlické hory Mts. Protected Landscape Area were also documented by VACEK et al. (2013). MÍCHAL (1983) also stated that mixed fir stands in near-natural forests have complicated structure and development dynamics. They often create richly structured multi-layered stands with multiple-generation structure (KORPEL 1995, JAWORSKI et al. 2000, SCHÜTZ 2001, JAWORSKI et al. 2002). This trend can be confirmed by the subsequent statistical evaluation of all four PRP where forest stands are characterized by higher species diversity and also by relatively high evenness. One of the characteristic features of forest stands is their species composition that expresses, as assumed by FORRESTER et al. (2013), the ability of species to cope with definite site conditions. Pronounced species differentiation of mixed fir forests is reported from Poland (JAWORSKI et al. 2006, DOBROWOLSKA, VEBLEN 2008) and from Bosnia and Herzegovina (MOTTA et al. 2011). From Slovakia from the Stará Pila locality, based on 29-year research, ŠTEFANČÍK (2006) demonstrated the current trend of a decrease in fir representation mainly at the cost of beech in the range of 4-36%. A decrease in the fir share at the cost of beech was also reported by ŠEBKOVÁ et al. (2011) from the area of the Boubín primeval forest based on 160-year research. A possibility of increasing the fir share in forests of central and southern Europe was described by KÖLLING (2007), who considered the fir as a species that could replace spruce in some warmer and drier localities. Nevertheless, the present representation of fir in forest stands confirms that its representation is not usually dominant although fir is the main spe-

cies from the aspect of natural species composition. KORPEL (1989) concluded that it may be the main codominant species especially in the 4th – 6th forest altitudinal zone (FAZ) while in the other FAZ it is only admixed or interspersed and usually has a lower vitality.

The actual standing volume of the main and secondary stand on the particular PRP is in the range of 237-598 m³.ha⁻¹. These values indicate pronounced variability of standing volumes on the PRP which is given especially by developmental stages of the particular forest stands and by past forest management interventions. If compared with standing volumes from Dolina Łopusznej locality in Gorczański NP, where the standing volume is up to 714 m³.ha⁻¹, (JAWORSKI et al. 2006), the standing volumes on the PRP we measured are substantially lower. Higher standing volume (800.3 m³.ha⁻¹) than on the PRP was reported by PANTIČ et al. (2011) from research plots in the Serbian Reserve Račanska Šljivovica.

The evaluation of the horizontal structure of overstorey trees indicates high variability on the particular plots. Mostly random distribution is on PRP 36 and 46, while the remaining plots show rather aggregated distribution. Different results were obtained by HOFMEISTER et al. (2008) on the basis of the spatial structure analysis and statistical evaluation of mixed fir stands in the Orlické hory Mts. that was done by means of Ripley's *K*-function; they stated that all developmental (growth) phases of spruce and fir showed aggregated distribution while the fir regeneration was positively correlated with the parent stand of spruce and the spruce regeneration was in a positive correlation with adult fir trees. From the Boubín primeval forest ŠEBKOVÁ et al. (2011) reported a random structure at the beginning of measurement that gradually changed into rather an aggregated structure which was explained by the mortality of fir trees. Similar results were obtained by JANIK et al. (2013), who revealed a trend of the mostly aggregated structure in mixed fir forests in the Western Carpathians.

It has been documented on the studied plots that hoofed game has a very negative or even destructive influence on fir natural regeneration. Natural regeneration of fir is in these forests without protection against game unsuccessful. These results are consistent with the results of BOTTERO et al. (2011) from mixed near-natural forests in the Lom Forest Reserve in Bosnia and Herzegovina, who observed a very high loss of fir mainly due to the negative influence of game. Similar conclusions were drawn by DIACI et al. (2010), who conducted a research on mixed stands of spruce, fir and beech in Slovenia and stated that the loss of fir is not caused only by devastation of natural regeneration by hoofed game but also by air pollution (SO₂ emissions) to which the fir is relatively vulnerable. CLOSSET-KOPP et al. (2006) considered increased vulnerability to frost and drought as another limiting factor of fir regeneration. It is similar to on PRPs in the Krkonoše Mts., where fir in these stands was dying rapidly during air pollution disturbance in the 80s of the 20th century. Based on our results and in accordance with similar studies from Germany and Switzerland (KENK, GUEHNE 2001; PUETTMANN et al. 2009) the health status and regeneration capacity of most

fir stand significantly improved during the last decades. In suitable light conditions and without game pressure high density of fir regeneration and its development are then ensured.

As demonstrated by PALUCH (2005), higher numbers of fir recruits occur in forests with a higher share of old trees, which is explained by a possible more intensive fall of seeds, therefore it is desirable to maintain the share of these old fertile fir trees as high as possible. It is crucial at the same time that the trees from lower diameter classes will continually reach higher classes and so the aging of forest stands will be prevented as reported from Slovenia by FICKO et al. (2011). GRASSI et al. (2004) believed that the existence of so called sapling bank was very important in forest stands where cultural practices for the support of fir are performed. Fir natural regeneration may gain a significant lead over the other tree species in this way, especially over beech and spruce (SZYMURA 2005).

5. Conclusion

The share of fir on the studied PRP is very different. A higher number of prospering recruits is only on the plots that have been protected by fencing for a long time (at present or in the past). Even so, after 30-year regeneration period the mean height of fir recruits is around 1.0 m showing slow initial height growth of fir and thus also longer period of high vulnerability of fir toward game damage. On the other hand, the ability of fir to survive in shade and to outcompete other species contributes to a height diversity of fir regrowth and subsequently to distinct diversification of whole forest stands.

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